FURONG JIA

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Durham, North Carolina, United States

Research Interests

I study **how data shapes LLM behavior** and **how LLMs interpret data**, toward AI that is **reliable, interpretable, and robust**. I examine how **training data** shapes model knowledge and reasoning, how to **evaluate** models faithfully, and how they behave under **distribution shift** across text, time series, and multimodal data. More recently, I study **LLM agents** for scientific discovery and **human–agent interaction**, with applications in **healthcare**.

Education

Duke University — *Durham, NC*
Ph.D. in Computer Science — Aug. 2024 – 2029 (expected)
Advisor: Monica Agrawal — GPA: 4.0/4.0
M.S. in Computer Science (en route) — Sep. 2026

University of Southern California — *Los Angeles, CA*
B.S. in Computer Science; B.S. in Applied and Computational Mathematics — May 2024
W.V.T. Rusch Engineering Honors Program; Viterbi Grand Challenges Scholar — GPA: 3.96/4.0

Research Experience

IBM Research, Core AI Team — *Boston, MA*
Research Intern; Mentor: Akash Srivastava — May 2026 – Aug. 2026

- **When Does the Harness Matter for LLM-Driven Discovery?** Led a comparison of 11 harnesses, from best-of-N sampling to evolutionary search and coding agents, across math, GPU-kernel, and LM-training tasks with Claude Opus 4.6 and Sonnet 4.5. Harness choice often mattered more than a model upgrade, but the winner depends on the setup. Counting the budget in API cost rather than evaluation count changed the leader in half of the settings, since harnesses differ over tenfold in cost per candidate. Running the same greedy-refinement loop with a coding agent instead of a fixed pipeline improved every task with Opus but only one with Sonnet. [12]

DukeNLP, Duke University — *Durham, NC*
Advisor: Monica Agrawal — Aug. 2024 – Present

- **How Faithfully do Language Models Acquire Probabilistic Knowledge?** Continually pretrained OLMo-2 1B and 7B on millions of notes from 650 synthetic Bayesian networks, so every learned probability can be checked exactly. Models learn individual probabilities closely but compress the relations between them, understating how far apart outcomes are and how much evidence should move a prediction. Because standard metrics miss this *relational compression*, probabilistic knowledge should be evaluated on relations, not single values. [11]
- **Interpreting Dataset Shift in Clinical Notes.** Built a benchmark that creates controlled shifts in real MIMIC-IV notes, and *SIReNs*, which explains a shift by showing an LLM representative notes from each dataset. SIReNs explains clear shifts nearly as well as an oracle, but gradual shifts spread thinly across many notes remain hard, motivating ongoing work on LLM agents for shifts with multiple causes. [3]
- **How does my language model understand clinical text?** Built *MedLingo*, a clinical-jargon benchmark from real hospital notes, and traced what open LLMs know to their pretraining corpora. Models understand jargon in rough proportion to its frequency there. Surprisingly, how often a term is used in real clinical notes barely predicts how often it appears in pretraining data, a gap between what LLMs learn and what clinical deployment needs. They also repeat disputed medical claims in the data, especially when a question presumes them. [4]
- **Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM.** Showed that a Naive Bayes ranker built only from concept–diagnosis counts in an LLM's pretraining corpus (or its open reproduction) diagnoses MedQA cases as accurately as the LLM itself (OLMo 7B, LLaMA 65B). Corpus statistics can thus explain much of the benchmark score, although the two methods get different questions right and work best combined. [2]

Melady Lab, University of Southern California — *Los Angeles, CA*
Advisor: Yan Liu — May 2023 – Jun. 2024

- **GPT4MTS: Prompt-Based Large Language Model for Multimodal Time-Series Forecasting.** Used LLMs to pair time series with summaries of related news, building a GDELT news dataset covering 55 U.S. regions. Adding the news text to GPT-2 as soft prompts cut forecasting error (MSE) by 4% over the best LLM baseline. [5]
- **TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting.** Co-developed and ran training experiments for a GPT-2 forecaster that splits each series into trend, seasonal, and residual parts and uses learned prompts to handle distribution shift. In zero-shot tests on unseen datasets it outperformed the baselines (e.g., 19% lower MAE than PatchTST on ETTm1). [6]

GLAMOR Lab, University of Southern California *Los Angeles, CA*
Advisor: Jesse Thomason Aug. 2022 – Jun. 2024

- **I2I: Initializing Adapters with Improvised Knowledge.** Trained CLIP-BART models for a continual-learning method that initializes each new task's adapter by distilling from earlier tasks' adapters. On five VQA tasks it transferred more knowledge than separate adapters (+4%), while AdapterFusion transferred less (−2%). [7]

Publications

Conference Publications *Equal contribution.

[1] **Batch-of-Thought: Cross-Instance Learning for Enhanced LLM Reasoning**
Xuan Yang, **Furong Jia**, Roy Xie, Xiong Xi, Hengwei Bian, Jian Li, Monica Agrawal.
EMNLP 2026 Main

[2] **Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM**
Furong Jia*, Yuan Pu*, Finn Guo, Monica Agrawal.
AACL-IJCNLP 2026 Main

[3] **Interpreting Dataset Shift in Clinical Notes**
Shariar Vaez-Ghaemi*, **Furong Jia***, Monica Agrawal.
ML4H 2025. PMLR 297:243–262, 2026.

[4] **How does my language model understand clinical text?**
Furong Jia, David Sontag, Monica Agrawal.
CHIL 2025. PMLR 287:720–743.

[5] **GPT4MTS: Prompt-Based Large Language Model for Multimodal Time-Series Forecasting**
Furong Jia, Kevin Wang, Yixiang Zheng, Defu Cao, Yan Liu.
EAAI 2024. Proc. AAAI 38(21):23343–23351.

[6] **TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting**
Defu Cao, **Furong Jia**, Sercan Ö. Arık, Tomas Pfister, Yixiang Zheng, Wen Ye, Yan Liu.
ICLR 2024.

[7] **I2I: Initializing Adapters with Improvised Knowledge**
Tejas Srinivasan, **Furong Jia**, Mohammad Rostami, Jesse Thomason.
CoLLAs 2023. PMLR 232:923–935.

Workshop / Findings & Abstracts

[8] **What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking**
Raymond M. Xiong, **Furong Jia**, Lionel Wong, Monica Agrawal.
ML4H 2025 Findings. arXiv:2601.15674, 2026.

[9] **Leveraging Large Language Models to Identify Barriers to Kidney Transplant in the STEPS Study**
Monica Agrawal, Metin Gurcan, Sarah Peskoe, Freya Gulamali, **Furong Jia**, Tara Strigo, Ebony Boulware, Clarissa Diamantidis.
Am. J. Kidney Dis. 87(5):S203–S204, 2026. (Abstract)

Preprints

[10] **Challenges of Auditing: Variability in Outputs of Large Language Models for Health**
Yuan Pu, Yewon Chang, **Furong Jia**, Xunjian Yin, Jessica Ma, Ayman Ali, Monica Agrawal.
arXiv:2609.16590, 2026.

Manuscripts Under Review

[11] **How Faithfully do Language Models Acquire Probabilistic Knowledge?**
Furong Jia, Xunjian Yin, Yuan Pu, Monica Agrawal.

[12] **When Does the Harness Matter for LLM-Driven Discovery?**
Furong Jia, Xinheng Ding, Chengrui Qu, Monica Agrawal, Akash Srivastava.

Honors and Awards

NeurIPS: Top Reviewer (2026).

University of Southern California: Albert Dorman Future Leader Award (2024); Student Recognition Awards (2024); Provost's Research Fellowship (Fall 2023); CURVE Research Fellowship (2022–2023); Academic Achievement Award (2021); ABC Innovation Prize (2021).

Service

- **Reviewer:** NeurIPS (2025, 2026); ICLR (2027); ML4H (2025, 2026); ACL ARR: Feb./Jun./Oct. 2024, Feb./May 2025; NeurIPS FMTS Workshop (2026); COLM DAIH Workshop (2026).
- **Teaching Assistant, Duke:** Natural Language Processing (Spring 2026); Applied Machine Learning (Spring 2025).
- **Teaching Assistant, USC:** Algorithms and Theory of Computing (Spring 2022); Web Publishing and Front-end Development (Fall 2021 – Spring 2024).